

BARNEY R. GILLIOM

QA Automation Engineer | Python & API Developer | Full-Stack Developer | DevOps
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PROFESSIONAL SUMMARY

SDGKU-trained Full Stack Developer and QA Automation Engineer with production experience building, testing, and deploying five independent real-world applications across e-commerce, REST APIs, system diagnostics, marketing platforms, and UI component libraries. Proficient in Python (unittest), JavaScript (Jest), Django REST APIs, React/TypeScript, PostgreSQL, FastAPI, and CI/CD automation via GitHub Actions. Delivers zero-defect, production-grade software — from thread-safe desktop applications with local HTTP APIs to cloud-deployed full-stack platforms — with a precision-first mindset forged in federal QA, military electronics, and FAA-certified air traffic control. All projects are independently developed, live production deployments; not academic exercises.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, PowerShell, Bash

Testing & QA: unittest (Python), Jest (JavaScript), API Testing, Manual Testing, End-to-End Testing, Test Case Design, Bug Reporting, Regression Testing

Backend & APIs: Django, Flask, FastAPI, Node.js, Express.js, RESTful API Design, CRUD Operations, JWT Authentication, Session Auth, Role-Based Access Control, Webhooks

Frontend & UI/UX: React, TypeScript, CustomTkinter, HTML5, CSS3, Vanilla JS, Responsive Design, Modal Behavior, Filter & Sort Logic, Keyboard Navigation, Dynamic UI

Desktop & Systems: Windows OS integration, WMI/CIM (PowerShell), Event Log parsing, UAC elevation, System tray (pystray), SQLite metrics, Matplotlib live charts

Databases: PostgreSQL, SQLite, Django ORM, SQL Schema Design, Database Migrations

DevOps & CI/CD: GitHub Actions, Git, GitHub, Render, Uvicorn, Gunicorn, WhiteNoise, Linux/Ubuntu CLI, PowerShell Scripting, PyInstaller (.exe builds)

AI Dev Tools: GitHub Copilot (VS Code), OpenAI Codex, Claude (Anthropic), Grok, Prompt Engineering

Security: CSRF Protection, Authentication & Authorization, Session Hardening, Input Validation, Bearer Token Auth, Access Control

Methodologies: Agile/Scrum, MVC Architecture, SDLC, Technical Documentation, Code Review, Modular Architecture

TECHNICAL PROJECTS — PRODUCTION DEPLOYMENTS

PC-Checker — Windows Health & Diagnostics Desktop Application | *Python, FastAPI, CustomTkinter, SQLite, PowerShell/WMI, Matplotlib, psutil, fpdf2, Render* | github.com/dallas8000-ops/PC-Checker

- Architected and built a product-shaped Windows desktop application — not a tutorial script — with multi-tab CustomTkinter GUI, background threading, UAC-aware administrator relaunch, optional system tray (pystray), and settings persisted to disk
- Implemented deep OS integration via PowerShell/WMI (CIM) for hardware identity, Event Log error pattern analysis, Defender/Windows Update/winget catalog inspection, and registry-backed disk hints
- Designed a thread-safe SharedState model feeding three independent consumers: the desktop GUI, a local FastAPI HTTP server (/api/v1/*), and a browser-based web dashboard — one data layer, multiple interfaces
- Built local REST API with FastAPI/uvicorn exposing live metrics, diagnostics, disk analysis, SQLite metric samples, and on-demand scan triggers; includes OpenAPI /docs and optional Bearer token webhook POST to cloud endpoint
- Implemented SQLite-backed background metrics sampling, live Matplotlib CPU/RAM trend charts, and structured JSON/HTML/PDF export pipeline via fpdf2

- Deployed optional read-only cloud mirror on Render (render_web/) that ingests JSON snapshots via webhook — demonstrating clear architectural judgment: WMI runs on Windows, cloud UI is a stateless viewer
- Packaged as standalone PCChecker.exe via PyInstaller build script; modular checks/ architecture separates memory, storage, hardware, events, and deep-system collectors for maintainability

Kristie Store — Full-Stack E-Commerce Platform | *Python, Django, React, TypeScript, PostgreSQL, GitHub Actions* | kristie-store.onrender.com

- Sole developer for all design, build, testing, and deployment of a live production e-commerce platform serving real customers
- Architected and validated RESTful APIs using Django with full CRUD operations, session-based authentication, and CSRF protection
- Managed PostgreSQL database with 80+ seeded products; wrote Python unittest suite covering cart logic, auth flows, and schema migrations with 100% passing rate
- Implemented advanced UI/UX: responsive product grid, dynamic cart updates, multi-criteria filtering, and SMTP email order confirmation
- Integrated WhatsApp order flow and executed end-to-end manual QA across full checkout and notification pipeline
- Configured GitHub Actions CI/CD pipeline for automated build validation and zero-downtime deployment to Render cloud platform

FrontLine Digital — Production Marketing & Business Platform | *React, TypeScript, Node.js, GitHub Actions, Render* | gilliomfrontlinedigital.onrender.com

- Built and deployed a production business marketing platform as sole developer with fully responsive React/TypeScript SPA architecture
- Implemented multi-page component structure with dynamic UI, professional layout, and content management sections
- Configured CI/CD via GitHub Actions — automated deployment on every push with zero manual deployment steps

Django REST Blog API — Portfolio & Content Platform | *Python, Django, PostgreSQL, JWT, Session Auth* | blog-2-hggg.onrender.com

- Built a production RESTful API with full CRUD, role-based access control, JWT + session authentication, and structured error handling
- Validated all authentication flows, permission logic, and API edge cases through Python unittest and manual API testing
- Implemented responsive multi-page front-end: navigation, About/Contact/Blog, project gallery, and tech stack showcase

React Product Catalog — UI/UX Component Library | *React, TypeScript, Jest, Vite* | github.com/dallas8000-ops/React-Store-Catalog

- Built modular React/TypeScript component library with Jest unit test suite covering rendering, interaction, and edge case behavior
- Implemented advanced UI features: modal behavior, multi-criteria filter logic, keyboard navigation, and ARIA-compliant accessible components
- Actively maintained with iterative feature updates and expanded test coverage — demonstrates continuous improvement discipline

PROFESSIONAL EXPERIENCE

QA Engineer / Full Stack Developer (Project-Based) | **San Diego Global Knowledge University** | Tampa, FL | 2025 – 2026

- Built, tested, and deployed five production applications as sole developer; executed full QA validation across the complete SDLC
- Wrote Python unittest and Jest test suites for API endpoints, authentication logic, UI components, and system integrations
- Configured and maintained GitHub Actions CI/CD pipelines with automated build, test, and deployment stages
- Documented test cases, bug reports, and deployment runbooks following Agile/Scrum workflow and sprint cadence
- Completed SDGKU Full Stack Software Development Technical Diploma — May 2026

Quality Assurance Analyst | Triple Canopy (Federal Contractor) | Kuwait | 2018

- Performed QA analysis on high-security threat-detection systems achieving 99.9% accuracy in mission-critical federal environments
- Executed diagnostics, system calibration, and validation protocols; produced formal technical QA documentation per regulatory standards
- Identified and coordinated corrective action on equipment faults in compliance with federal compliance frameworks

Law Enforcement Officer | U.S. Army Military Police | Waco PD | Dallas PD | Texas / Washington | 1998 – 2015

- Maintained rigorous investigative documentation and chain-of-custody records under strict federal and state compliance standards
- Operated law enforcement databases and communication systems under high-pressure, time-critical operational conditions
- Attained TCOLE Master Police Officer Certification; received commendations for precision and judgment across multiple jurisdictions

Air Traffic Controller / Electronics Technician | United States Army | U.S. / Overseas | 1987 – 1999

- FAA-Certified ATC operator; managed safe aircraft sequencing requiring split-second accuracy and zero-defect execution
- Diagnosed and repaired communication, radar, and navigation systems using multimeters, oscilloscopes, and signal analyzers
- Maintained precise technical logs and system documentation across mission-critical operational environments

EDUCATION & CERTIFICATIONS

Full Stack Software Development — Technical Diploma | San Diego Global Knowledge University | Tampa, FL | 2025 – 2026

Python, React, TypeScript, Node.js, PostgreSQL, GitHub Actions, Agile/Scrum | Completed May 2026

Information Technology — Technical Diploma | MyComputerCareer | 2024 – 2025

IT Security, CompTIA, Networking, Systems Administration

Associate of Arts — Business Administration | American InterContinental University | 2007 – 2008

Associate of Arts — Legal Assistant/Paralegal | Northwest Mississippi Community College | 1980 – 1982

Certifications:

- AI-900: Microsoft Azure AI Fundamentals | IT Security Certification (CompTIA)
- FAA Air Traffic Controller Certification | FAA Helicopter Communication Systems/Repair
- TCOLE Master Police Officer Certification | Immersive Full Stack Software Development (SDGKU)